KAVYA KOLICHELAM

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PROFESSIONAL SUMMARY

MS Computer Science (Data Science) graduate with collaborative faculty research in hierarchical deep learning and neuro-symbolic AI, plus 2+ years of experience in data analysis, machine learning, and AI-driven solution development. Skilled in Python, SQL, and end-to-end ML workflows — from data cleaning and EDA through model building, evaluation, and deployment. Hands-on experience with deep learning, NLP, LLMs, RAG, and Generative AI. Proven ability to build scalable data pipelines and dashboards, communicate insights to both technical and non-technical stakeholders, and deliver impactful results across large-scale datasets (200K+ records).

TECHNICAL SKILLS

Languages & Querying: Python, SQL, Java, C, C++

Machine Learning: Regression, Classification, Ensemble Methods (XGBoost, Random Forest), Feature Engineering, Model Evaluation, Cross-Validation, Scikit-learn, Probabilistic ILP

Deep Learning & CV: PyTorch, TensorFlow, CNN, MobileNetV2, SegFormer, U-Net++, Transfer Learning, Domain Adaptation, Multi-Task Learning

NLP & Generative AI: TF-IDF, Sentiment Analysis, NLTK, spaCy, HuggingFace Transformers, DistilBERT, LLMs, RAG, Prompt Engineering, OpenAI APIs

Data Analysis & Statistics: EDA, Statistical Analysis, Data Mining, Predictive Analysis, Trend Analysis, A/B Testing Concepts, Hypothesis Testing, Calibration Analysis (ECE)

Data Processing: Pandas, NumPy, SciPy, Data Cleaning, Data Validation, Data Wrangling, ETL Concepts, Feature Engineering

Visualization & BI: Power BI, Tableau, Matplotlib, Seaborn, NetworkX, WordCloud, Excel (Pivot Tables, VLOOKUP, XLOOKUP, Power Query), Google Sheets

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Google Colab, MS Office Suite

PROFESSIONAL EXPERIENCE

Teaching Assistant – Machine Learning | Lawrence Technological University, Southfield, MI *Aug 2025 – May 2026*

- Guided 40+ students through ML model development, data preprocessing, feature engineering, and predictive modeling in a fast-paced academic environment
- Validated analytical outputs and student-built data pipelines; identified processing errors and guided corrective actions to improve result accuracy
- Communicated complex ML, statistical, and data concepts clearly to students with varying technical backgrounds, reinforcing analytical best practices
- Assisted with technical documentation, reporting, and cross-team coordination to meet semester deadlines

Machine Learning Engineer | Start Tech Innovations, India *Jan 2023 – Dec 2023*

- Developed and deployed regression and classification models using Python, Scikit-learn, and XGBoost for production business workflows across structured datasets
- Performed extensive EDA, feature engineering, and statistical analysis to identify trends, anomalies, and optimization opportunities
- Evaluated models using cross-validation, precision, recall, ROC-AUC, and RMSE; iterated systematically to improve production performance
- Designed and automated scalable ML data pipelines, reducing manual processing time and ensuring reliable, repeatable outputs
- Collaborated with cross-functional teams to translate analytical findings into actionable recommendations for business stakeholders

Data Analyst Intern | Tech Solutions Pvt Ltd, Hyderabad, India *May 2022 – Dec 2022*

- Analyzed large datasets using SQL and Python (Pandas, NumPy) to extract insights, identify trends, and support data-driven decision-making across departments
- Performed statistical analysis, data mining, and predictive analysis to uncover optimization opportunities and surface data quality issues

- Built and maintained Power BI dashboards and automated reports to enable self-service analytics for non-technical stakeholders
- Cleaned, validated, and reprocessed Excel and CSV datasets; investigated discrepancies and implemented fixes improving downstream reporting accuracy
- Collaborated with cross-functional teams to deliver insights supporting operational efficiency and KPI performance tracking

PROJECT EXPERIENCE

Neuro-Symbolic Hybrid Classification System | *Python*, *Scikit-learn*, *TF-IDF*, *Probabilistic ILP*, *Multinomial Naive Bayes*, *NetworkX*

- Designed two-tier hybrid AI architecture combining Probabilistic Inductive Logic Programming (P-ILP) rules with calibrated Multinomial Naive Bayes for interpretable text classification on 209,527 HuffPost news articles across 10 categories
- Achieved 82.85% overall accuracy with 24.3% of decisions made through transparent rule-based logic and 75.7% through calibrated ML; Expected Calibration Error of 1.4% (target <5%) validates confidence score reliability
- Implemented and compared four fusion strategies (Confidence Routing, Weighted Voting, Cascaded Fusion, Stacking Ensemble); achieved 91.3% rule-ML fidelity confirming ante-hoc interpretability
- Documented as Collaborative Research Project formal report covering literature review, methodology, hypothesis validation, and limitations

Social Network Sentiment Analysis for Mental Health Awareness | *Python*, *NLP*, *HuggingFace Transformers*, *DistilBERT*, *NLTK*, *Scikit-learn*

- Built end-to-end NLP pipeline classifying sentiment across 38,000+ Reddit posts using VADER for three-class rule-based labeling and DistilBERT for transformer-based binary classification
- Applied K-means thematic clustering (k=5) on TF-IDF features with PCA dimensionality reduction; combined with WordCloud and temporal trend analysis to surface latent themes for stakeholder reporting
- Co-authored project: handled imbalanced classes, produced clear visualizations using Matplotlib and Seaborn, and communicated findings to non-technical audiences through structured visual reports

Hierarchical Coarse-to-Fine Crack Detection Framework | **Collaborative Research Project with Dr. Wisam Bukaita** | *PyTorch*, *SegFormer-B2*, *MobileNetV2*, *U-Net++*, *Transfer Learning*, *Domain Adaptation*

- Designed and implemented a three-stage deep learning framework for automated concrete structural inspection as part of a collaborative research project with Dr. Wisam Bukaita: MobileNetV2 classifier (91.68% accuracy, ROC-AUC 0.934), SegFormer-B2 segmentation at 384×384 with TTA (Dice 0.7572, IoU 0.6353), and geometry stage estimating crack length and width from masks
- Benchmarked 10+ architectures including U-Net variants, Attention U-Net, DeepLabV3+, U-Net++ ResNet34, EfficientNet-B4 U-Net++, and SegFormer-B0/B2 before selecting SegFormer-B2 as the final segmentation model
- Implemented two-stage domain adaptation on the DeepCrack dataset, improving cross-domain Dice to 0.6553; designed pipeline-level reliability metrics with 85% of images filtered before segmentation
- Conducted statistical validation, ablation studies, and computational efficiency analysis; documented results in formal research manuscript

LLM-Based AI Assistant with Retrieval-Augmented Generation | *OpenAI APIs*, *RAG*, *Prompt Engineering*, *Python*, *Vector Retrieval*

- Designed and developed a Retrieval-Augmented Generation (RAG) chatbot using OpenAI APIs, implementing structured document indexing, semantic retrieval, and context-aware prompt engineering
- Integrated dynamic retrieval pipeline to ground LLM responses in relevant source documents, improving factual accuracy and reducing hallucinations
- Applied advanced prompt engineering techniques including chain-of-thought prompting and few-shot examples to optimize response quality across diverse query types

EDUCATION

Master of Science – Computer Science (Data Science) GPA: 3.8
Lawrence Technological University, Southfield, MI *May 2026*

Bachelor of Technology – Electrical & Electronics Engineering
Bhoj Reddy Engineering College for Women, Hyderabad, India *2019 – 2023*

CERTIFICATIONS

- Machine Learning – Coursera (Andrew Ng)
- Deep Learning Specialization – Coursera
- Generative AI Fundamentals – Coursera
- Data Science Tools – IBM
- Python for Data Science – IBM
- SQL for Data Science – Coursera
- Data Analysis with Python – IBM / Coursera
- Data Visualization with Python – IBM / Coursera
- Excel Basics for Data Analysis – IBM / Coursera
- Power BI for Beginners – Microsoft / Coursera